Question and design

Does model width improve contact accuracy at the same one-epoch exposure? Baseline: exp277, native plus MPNN corpus, 24 layers. Hidden width 2048 -> 4096; MLP width 8192 -> 16384. Keep 32 query / 8 KV heads; head width 64 -> 128. Keep data, optimizer schedule, seeds and global batch fixed.

Architecture and smoke verified

5,861,984,256 parameters, versus exp277's 1,471,374,336 (3.984x). 12 local configuration/tokenizer/data checks passed. 64-H100 smoke: ten updates, train loss 8.56948, eval loss 8.03575. 2.821 seconds/update; native and HF checkpoints saved with tokenizer. Compressed smoke schedule was spiky; this is an execution check.

Current status

GPU reached step 8438 (3.17%); train loss 3.158, validation loss 3.465. Checkpoint barrier timeout stopped training; complete step 8120 remains. Scheduler falsely reported success; runtime now verifies final checkpoint. 32-H100 restart in startup; checkpoint restoration not yet confirmed. TPU trial moved to v5p-64 (32 chips); v6e cancelled before training. v5p offers more memory/bandwidth; twenty native-only updates planned. Placement and TPU throughput remain unconfirmed. Production migration: 320.4 GB; transfer approval still pending. Contact-accuracy results are pending.